SZABIST
University

Shaheed Zulfikar Ali Bhutto
Institute of Science and Technology

ROBOTICS & ARTIFICIAL INTELLIGENCE DEPARTMENT

Total Marks:_____

Obtained Marks: _____

Computing Vision

Lab Task # 08

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Code Example 1: Texture Analysis for Surface Defect Detection

A manufacturing company wants to automatically inspect metal surface sheets for defects like scratches, dents, or corrosion. These defects often manifest as irregular texture patterns on the surface.

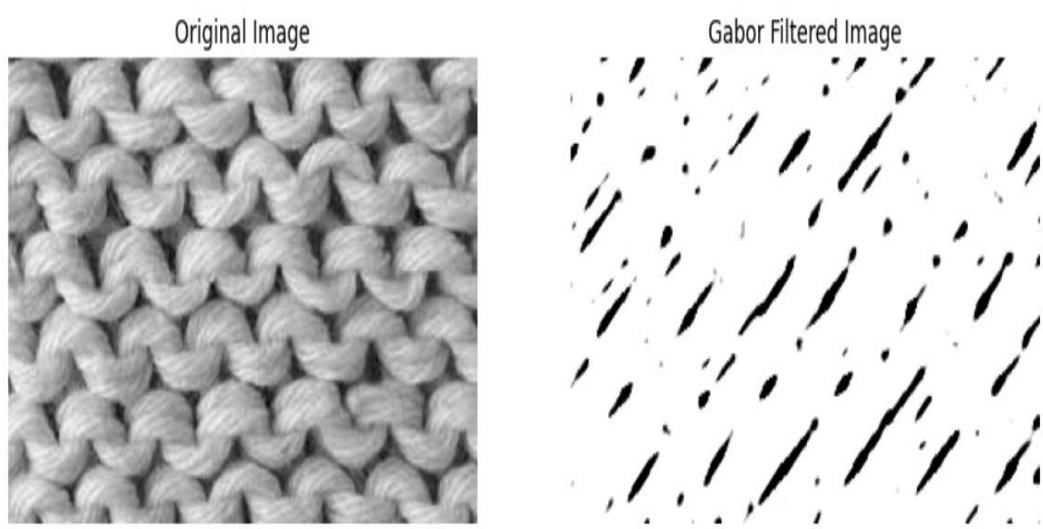
Task: Implement a texture analysis system using Gabor feature extraction that enhances texture features of metal surface images. The enhanced features can then be used for defect detection algorithms.

Steps:

- **Upload metal surface images captured from the production line.**
- **Apply Gabor filters at different orientations and scales to extract texture features.**
- **Highlight areas with unusual texture changes that may indicate defects.**

Application: Automated quality control in manufacturing and materials inspection.

```python
# Code Example 1: Texture Analysis for Surface Defect Detection
import cv2
import numpy as np
import matplotlib.pyplot as plt
gray = cv2.imread("D:/23108109/F24.jpg", cv2.IMREAD_GRAYSCALE)
kernel = cv2.getGaborKernel(ksize=(21, 21), sigma=8.0, theta=np.pi/4, lambd=10.0, gamma=0.5, psi=0, ktype=cv2.CV_32F)
filtered = cv2.filter2D(gray, cv2.CV_8UC3, kernel)
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1)
plt.imshow(gray, cmap='gray')
plt.title('Original Image')
plt.axis('off')
plt.subplot(1, 2, 2)
plt.imshow(filtered, cmap='gray')
plt.title('Gabor Filtered Image')
plt.axis('off')
plt.show()
```



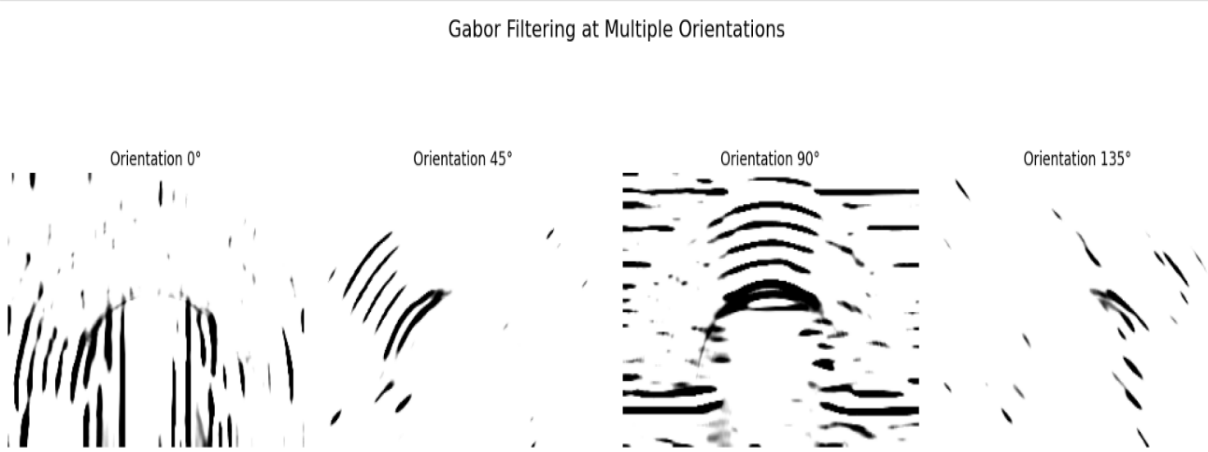
Code Example 2: Automated Architectural Surface Texture Analysis Using Gabor Filters
An architecture firm is working on a heritage building restoration project. They need to analyze high-resolution images of brick walls and tile surfaces to study the texture patterns, detect surface wear, cracks, and inconsistencies that are not easily visible to the naked eye.
Objective: Use Gabor filters at multiple orientations to extract detailed directional texture features from images of brick walls and tiles. This helps in:

- **Highlighting repetitive brick or tile patterns.**
- **Detecting irregularities such as cracks, erosion, or missing tiles.**
- **Quantifying texture variations for condition assessment.**

Task:

- **Upload high-resolution images of brick walls or tiled surfaces.**
- **Apply Gabor filters at orientations of 0°, 45°, 90°, and 135° to enhance directional features.**
- **Analyze the filtered images to identify areas of surface damage or texture anomalies.**
- **Support restoration planning by mapping damaged areas based on texture deviations.**

```python
# Code Example 2: Automated Architectural Surface Texture Analysis Using Gabor Filters
import cv2
import numpy as np
import matplotlib.pyplot as plt
gray = cv2.imread("D:/23108109/F25.jpg", cv2.IMREAD_GRAYSCALE)
orientations = [0, np.pi/4, np.pi/2, 3*np.pi/4]
plt.figure(figsize=(16, 5))
for i, theta in enumerate(orientations):
    kernel = cv2.getGaborKernel((21, 21), 4.0, theta, 10.0, 0.5, 0, ktype=cv2.CV_32F)
    filtered_img = cv2.filter2D(gray, cv2.CV_8UC3, kernel)
    plt.subplot(1, 4, i+1)
    plt.imshow(filtered_img, cmap='gray')
    plt.title(f"Orientation {int(np.degrees(theta))}°")
    plt.axis('off')
plt.suptitle('Gabor Filtering at Multiple Orientations', fontsize=16)
plt.tight_layout(rect=[0, 0, 1, 0.95])
plt.show()
```



Code Example 3: Fingerprint Ridge Enhancement Using Gabor Filters for Minutiae Detection

A forensic lab is developing an automated fingerprint recognition system. Fingerprint images often contain noise and inconsistent ridge clarity due to skin conditions or capture device limitations. To improve the accuracy of minutiae detection (ridge endings and bifurcations), the lab applies Gabor filters to enhance ridge structures by emphasizing their directional patterns.

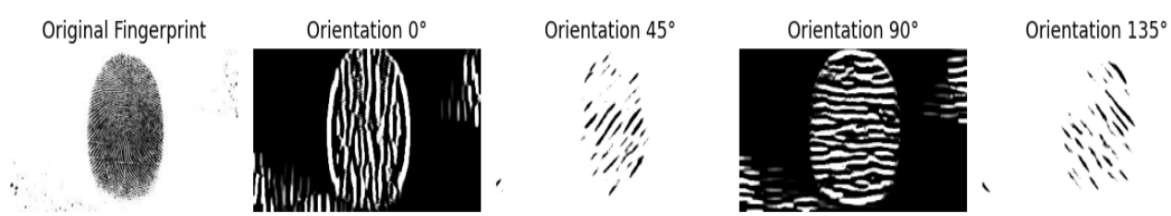
Objective:

- **Load fingerprint images.**
- **Apply Gabor filtering tuned to the ridge frequency and orientation.**
- **Enhance ridge clarity for improved feature extraction.**
- **Prepare the enhanced images for downstream minutiae detection algorithms.**

Benefits:

- **Enhanced ridge patterns improve minutiae extraction accuracy.**
- **Noise reduction and texture enhancement make fingerprint matching more reliable.**

```python
# Code Example 3: Fingerprint Ridge Enhancement Using Gabor Filters for Minutiae Detection
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread("D:/23108109/F26.jpg", cv2.IMREAD_GRAYSCALE)
if img is None:
    print("Error loading image.")
else:
    kernel_size = 21
    sigma = 5.0
    lambd = 10.0
    gamma = 0.5
    psi = 0
    orientations = [0, np.pi/4, np.pi/2, 3*np.pi/4]
    filtered_imgs = []
    plt.figure(figsize=(12, 6))
    plt.subplot(1, len(orientations)+1, 1)
    plt.title("Original Fingerprint")
    plt.imshow(img, cmap='gray')
    plt.axis('off')
    for i, theta in enumerate(orientations):
        kernel = cv2.getGaborKernel((kernel_size, kernel_size), sigma, theta, lambd, gamma, psi, ktype=cv2.CV_32F)
        filtered_img = cv2.filter2D(img, cv2.CV_8UC3, kernel)
        filtered_imgs.append(filtered_img)
        plt.subplot(1, len(orientations)+1, i+2)
        plt.title(f"Orientation {int(np.degrees(theta))}°")
        plt.imshow(filtered_img, cmap='gray')
        plt.axis('off')
    plt.tight_layout()
    plt.show()
```



Code Example 4: Texture Classification for Industrial Quality Inspection Using Gabor Features

A manufacturing company produces ceramic tiles and fabric materials. To maintain product quality, the company needs an automated system to classify surface textures to detect defects such as cracks, irregular patterns, or inconsistencies. Using Gabor filters, the system extracts texture features representing spatial frequencies and orientations, enabling effective classification of normal versus defective textures.

Objective:

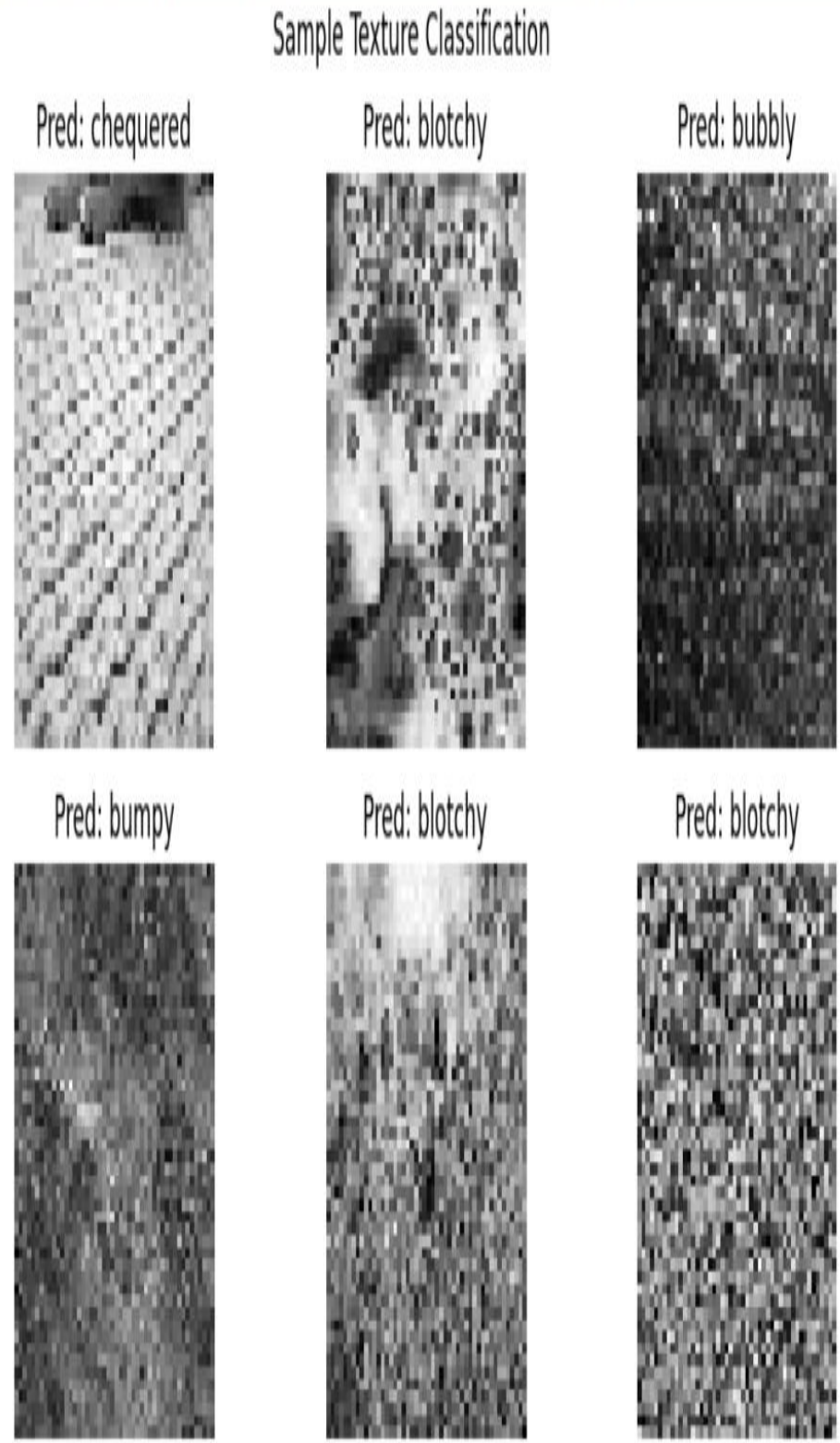
- **Load images of various textures (e.g., fabric, tiles).**
- **Apply Gabor filters at multiple orientations and scales to extract texture features.**
- **Use extracted features to classify the surface textures.**
- **Support quality control by automatically detecting anomalies or classifying textures.**

```python
# Code Example 4: Texture Classification for Industrial Quality Inspection Using Gabor Features
import cv2
import numpy as np
import matplotlib.pyplot as plt
import os
from sklearn.svm import SVC
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report, accuracy_score
def gabor_features(img, kernels):
    features = []
    img = img.astype(np.float32)
    for kernel in kernels:
        filtered = cv2.filter2D(img, cv2.CV_32F, kernel)
        features.append(filtered.mean())
        features.append(filtered.var())
    return features
dataset_path = "D:/23108109/dtd/images"
kernel_size = 15
sigmas = [4.0]
lambdas = [10.0]
orientations = [0, np.pi/2]
kernels = [
    cv2.getGaborKernel(
        (kernel_size, kernel_size),
        sigma,
        theta,
        lambd,
        0.5,
        0,
        ktype=cv2.CV_32F
    )
    for sigma in sigmas
    for lambd in lambdas
    for theta in orientations
]
X, y, images = [], [], []
label_map = {}
label_id = 0
for class_name in os.listdir(dataset_path)[:6]:
```

```python
    class_path = os.path.join(dataset_path, class_name)
    if os.path.isdir(class_path):
        label_map[label_id] = class_name
        for img_name in os.listdir(class_path)[:10]:
            img_path = os.path.join(class_path, img_name)
            img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
            if img is not None:
                img = cv2.resize(img, (48, 48))
                features = gabor_features(img, kernels)
                X.append(features)
                y.append(label_id)
                images.append(img)
        label_id += 1
X = np.array(X)
y = np.array(y)
print("Total samples:", len(X))
X_train, X_test, y_train, y_test, img_train, img_test = train_test_split(
    X, y, images,
    test_size=0.25,
    random_state=42,
    stratify=y
)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
clf = SVC(kernel='rbf', C=1.0, gamma='scale')
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)
print("\nClassification Report:\n")
labels = np.unique(y_test)
print(classification_report(
    y_test,
    y_pred,
    labels=labels,
    target_names=[label_map[i] for i in labels]
))
print(f"\nAccuracy: {accuracy_score(y_test, y_pred)*100:.2f}%")
plt.figure(figsize=(10, 5))
for i in range(min(6, len(img_test))):
    plt.subplot(2, 3, i + 1)
```

```python
for i in range(min(6, len(img_test))):
    plt.subplot(2, 3, i + 1)
    plt.imshow(img_test[i], cmap='gray')
    plt.title(f"Pred: {label_map[y_pred[i]]}")
    plt.axis('off')
plt.suptitle("Sample Texture Classification")
plt.show()
```

```
Total samples: 60

Classification Report:

              precision    recall  f1-score   support

      banded       0.40      1.00      0.57         2
     blotchy       0.20      0.50      0.29         2
     braided       0.00      0.00      0.00         3
      bubbly       0.00      0.00      0.00         3
       bumpy       0.00      0.00      0.00         3
   chequered       0.00      0.00      0.00         2

    accuracy                           0.20        15
   macro avg       0.10      0.25      0.14        15
weighted avg       0.08      0.20      0.11        15


Accuracy: 20.00%
```


Sample Texture Classification
Pred: chequered
Pred: blotchy
Pred: bubbly
Pred: bumpy
Pred: blotchy
Pred: blotchy

1. Fingerprint Ridge Enhancement for Forensic Applications

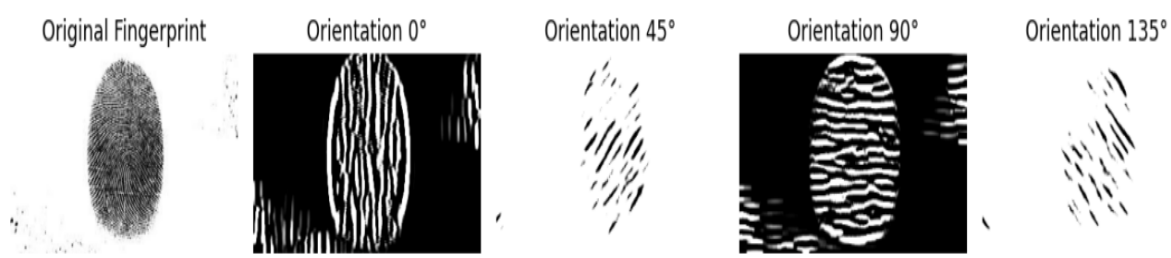
A forensic department is digitizing paper fingerprint records for building a centralized criminal database. However, many scans are faded or noisy, making it difficult to detect minutiae points (ridge endings and bifurcations).

Objective: Use Gabor filters to enhance fingerprint ridge patterns by tuning frequency and orientation parameters. This preprocessing step improves ridge visibility and supports more accurate feature extraction for identification.

Application Areas:

- **Criminal record matching**
- **Biometric authentication systems**
- **Law enforcement archives**

```python
# Code Example 3: Fingerprint Ridge Enhancement Using Gabor Filters for Minutiae Detection
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread("D:/23108109/F26.jpg", cv2.IMREAD_GRAYSCALE)
if img is None:
    print("Error loading image.")
else:
    kernel_size = 21
    sigma = 5.0
    lambd = 10.0
    gamma = 0.5
    psi = 0
    orientations = [0, np.pi/4, np.pi/2, 3*np.pi/4]
    filtered_imgs = []
    plt.figure(figsize=(12, 6))
    plt.subplot(1, len(orientations)+1, 1)
    plt.title("Original Fingerprint")
    plt.imshow(img, cmap='gray')
    plt.axis('off')
    for i, theta in enumerate(orientations):
        kernel = cv2.getGaborKernel((kernel_size, kernel_size), sigma, theta, lambd, gamma, psi, ktype=cv2.CV_32F)
        filtered_img = cv2.filter2D(img, cv2.CV_8UC3, kernel)
        filtered_imgs.append(filtered_img)
        plt.subplot(1, len(orientations)+1, i+2)
        plt.title(f"Orientation {int(np.degrees(theta))}°")
        plt.imshow(filtered_img, cmap='gray')
        plt.axis('off')
    plt.tight_layout()
    plt.show()
```



2. Texture Classification in Industrial Quality Control

A tile manufacturing unit needs an automated system to classify and inspect tiles for texture defects. Variations in the surface such as cracks, uneven patterns, or printing errors must be detected in real-time.

Objective: Extract Gabor features from tile surface images to represent orientation and frequency-based texture information. Use these features to train a machine learning model that classifies defective vs. acceptable tiles.

Application Areas:

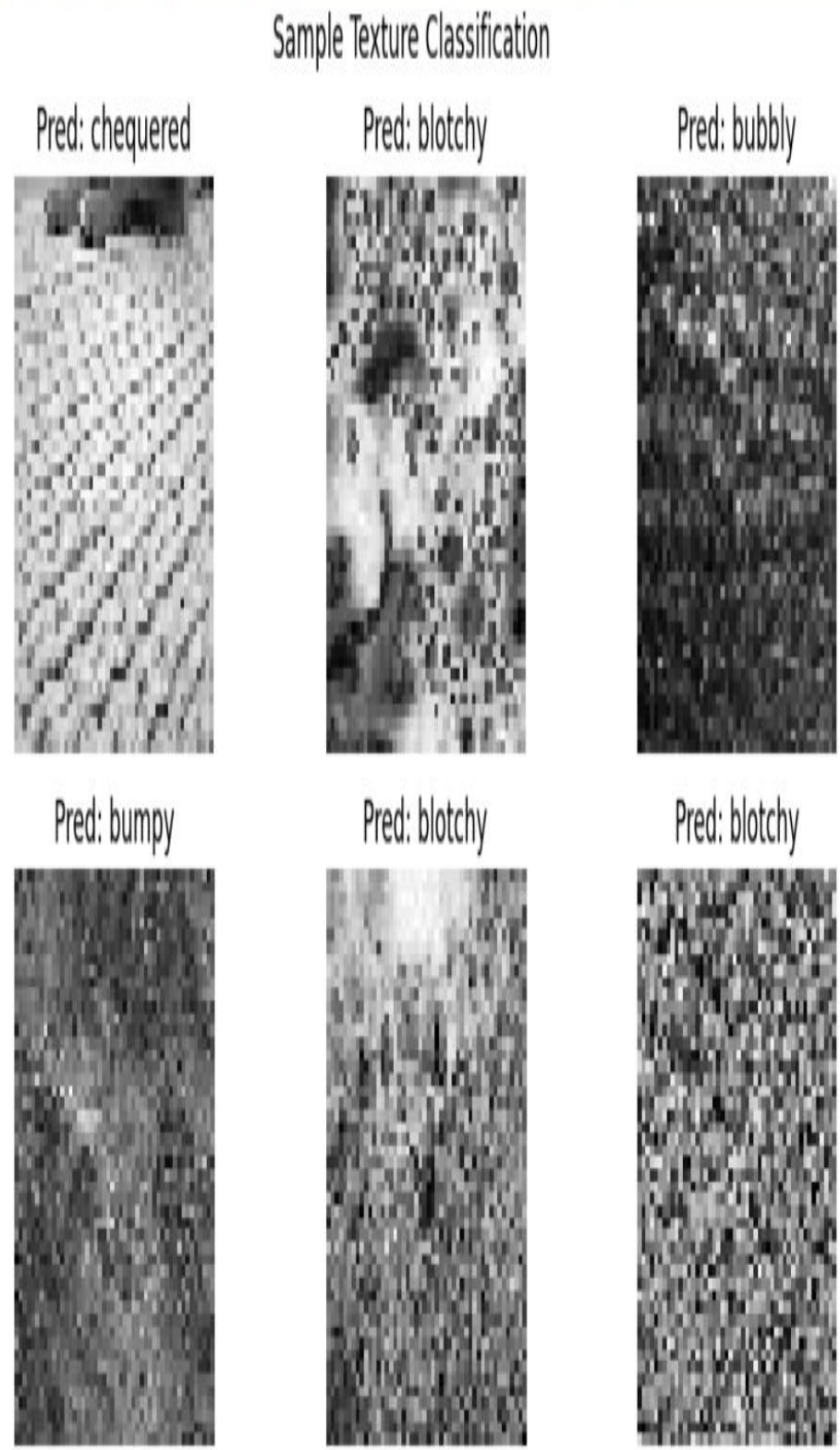
- **Tile and fabric quality control**
- **Surface inspection systems**
- **Automated defect detection**

```python
# Code Example 4: Texture Classification for Industrial Quality Inspection Using Gabor Features
import cv2
import numpy as np
import matplotlib.pyplot as plt
import os
from sklearn.svm import SVC
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report, accuracy_score
def gabor_features(img, kernels):
    features = []
    img = img.astype(np.float32)
    for kernel in kernels:
        filtered = cv2.filter2D(img, cv2.CV_32F, kernel)
        features.append(filtered.mean())
        features.append(filtered.var())
    return features
dataset_path = "D:/23108109/dtd/images"
kernel_size = 15
sigmas = [4.0]
lambdas = [10.0]
orientations = [0, np.pi/2]
kernels = [
    cv2.getGaborKernel(
        (kernel_size, kernel_size),
        sigma,
        theta,
        lambd,
        0.5,
        0,
        ktype=cv2.CV_32F
    )
    for sigma in sigmas
    for lambd in lambdas
    for theta in orientations
]
X, y, images = [], [], []
label_map = {}
label_id = 0
for class_name in os.listdir(dataset_path)[:6]:
```

```python
    class_path = os.path.join(dataset_path, class_name)
    if os.path.isdir(class_path):
        label_map[label_id] = class_name
        for img_name in os.listdir(class_path)[:10]:
            img_path = os.path.join(class_path, img_name)
            img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
            if img is not None:
                img = cv2.resize(img, (48, 48))
                features = gabor_features(img, kernels)
                X.append(features)
                y.append(label_id)
                images.append(img)
        label_id += 1
X = np.array(X)
y = np.array(y)
print("Total samples:", len(X))
X_train, X_test, y_train, y_test, img_train, img_test = train_test_split(
    X, y, images,
    test_size=0.25,
    random_state=42,
    stratify=y
)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
clf = SVC(kernel='rbf', C=1.0, gamma='scale')
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)
print("\nClassification Report:\n")
labels = np.unique(y_test)
print(classification_report(
    y_test,
    y_pred,
    labels=labels,
    target_names=[label_map[i] for i in labels]
))
print(f"\nAccuracy: {accuracy_score(y_test, y_pred)*100:.2f}%")
plt.figure(figsize=(10, 5))
for i in range(min(6, len(img_test))):
    plt.subplot(2, 3, i + 1)
```

```python
for i in range(min(6, len(img_test))):
    plt.subplot(2, 3, i + 1)
    plt.imshow(img_test[i], cmap='gray')
    plt.title(f"Pred: {label_map[y_pred[i]]}")
    plt.axis('off')
plt.suptitle("Sample Texture Classification")
plt.show()
```

```
Total samples: 60

Classification Report:

              precision    recall  f1-score   support

      banded       0.40      1.00      0.57         2
     blotchy       0.20      0.50      0.29         2
     braided       0.00      0.00      0.00         3
      bubbly       0.00      0.00      0.00         3
       bumpy       0.00      0.00      0.00         3
   chequered       0.00      0.00      0.00         2

    accuracy                           0.20        15
   macro avg       0.10      0.25      0.14        15
weighted avg       0.08      0.20      0.11        15


Accuracy: 20.00%
```


Sample Texture Classification
Pred: chequered
Pred: blotchy
Pred: bubbly
Pred: bumpy
Pred: blotchy
Pred: blotchy

3. Facial Expression Recognition for Mental Health Analysis

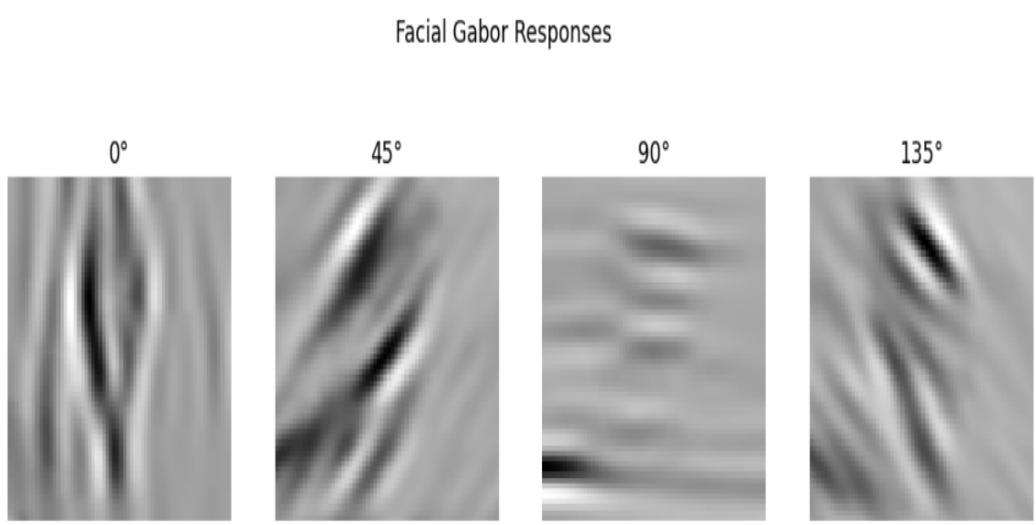
A health tech startup is building an application to assess emotional states (happy, sad, angry, etc.) using facial expressions captured from webcam feeds.

Objective: Apply Gabor filters to extract texture-based facial features that emphasize key regions (e.g., eyes, eyebrows, mouth). These features are later used in training classifiers for emotion recognition.

Application Areas:

- **Assistive technology for autism**
- **Virtual therapy sessions**
- **Behavioral monitoring tools**

```python
# 3. Facial Expression Recognition for Mental Health Analysis
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread("D:/23108109/F16.jpg", cv2.IMREAD_GRAYSCALE)
img = cv2.resize(img, (64, 64))
orientations = [0, np.pi/4, np.pi/2, 3*np.pi/4]
plt.figure(figsize=(12,4))
features = []
for i, theta in enumerate(orientations):
    kernel = cv2.getGaborKernel((21,21), 4.0, theta, 10.0, 0.5, 0, ktype=cv2.CV_32F)
    filtered = cv2.filter2D(img, cv2.CV_32F, kernel)
    features.append(filtered.mean())
    features.append(filtered.var())
    plt.subplot(1,4,i+1)
    plt.imshow(filtered, cmap='gray')
    plt.title(f"{int(np.degrees(theta))}°")
    plt.axis('off')
plt.suptitle("Facial Gabor Responses")
plt.show()
print("Feature Vector:", features)
```




```
Feature Vector: [np.float32(1398.2098), np.float32(365357.2), np.float32(2781.5586), np.float32(285335.8), np.float32(1403.3881), np.float32(707019.4),
```

4. Texture-Based Segmentation of Satellite Images

An environmental research institute wants to classify land types (forests, water bodies, urban regions) from satellite images based on surface texture patterns.

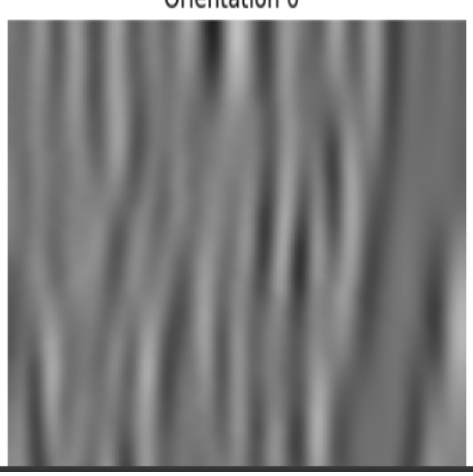
Objective: Use Gabor filters to extract local frequency features from satellite images, enabling better segmentation and classification of terrain types.

Application Areas:

- **Land cover classification**
- **Agriculture and irrigation planning**
- **Urban sprawl analysis**

```python
# 4. Texture-Based Segmentation of Satellite Images
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread("D:/23108109/F7.jpg", cv2.IMREAD_GRAYSCALE)
img = cv2.resize(img, (128, 128))
orientations = [0, np.pi/4, np.pi/2]
responses = []
for theta in orientations:
    kernel = cv2.getGaborKernel((31,31), 5.0, theta, 12.0, 0.5, 0)
    filtered = cv2.filter2D(img, cv2.CV_32F, kernel)
    responses.append(filtered)
    plt.imshow(filtered, cmap='gray')
    plt.title(f"Orientation {int(np.degrees(theta))}°")
    plt.axis('off')
    plt.show()
avg_response = [np.mean(r) for r in responses]
print("Texture Response Values:", avg_response)
if max(avg_response) == avg_response[0]:
    print("Detected: Water Region")
elif max(avg_response) == avg_response[1]:
    print("Detected: Forest Region")
else:
    print("Detected: Urban Region")
```

Orientation 0°



5. Architectural Material Recognition in Restoration Projects

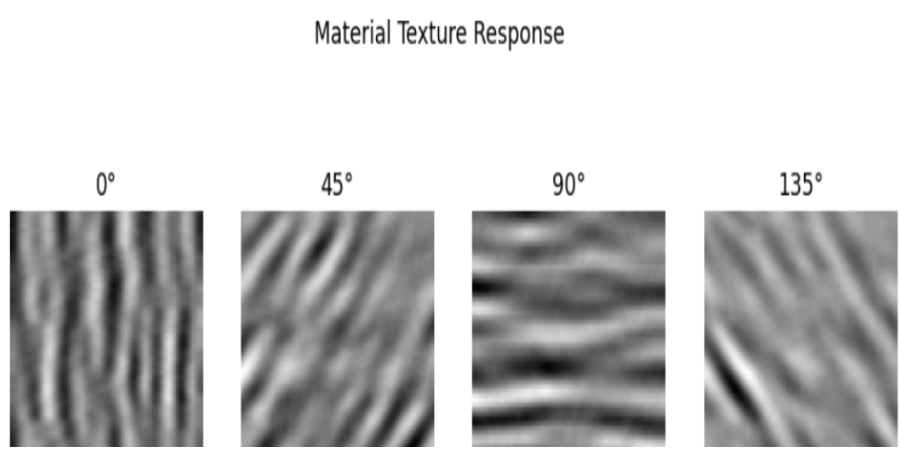
An architecture firm is analyzing old structures to determine the type and condition of building materials (e.g., different brick types, wood, plaster) for digital documentation and restoration.

Objective: Apply Gabor feature extraction to high-resolution images of wall surfaces to capture distinctive texture patterns. These features aid in automated material classification and degradation detection.

Application Areas:

- **Cultural heritage digitization**
- **Smart construction**
- **AR/VR-based architecture tools**

```python
# 5. Architectural Material Recognition in Restoration Projects
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread("D:/23108109/F27.jpg", cv2.IMREAD_GRAYSCALE)
img = cv2.resize(img, (64, 64))
orientations = [0, np.pi/4, np.pi/2, 3*np.pi/4]
features = []
plt.figure(figsize=(10,4))
for i, theta in enumerate(orientations):
    kernel = cv2.getGaborKernel((21,21), 6.0, theta, 10.0, 0.5, 0)
    filtered = cv2.filter2D(img, cv2.CV_32F, kernel)
    features.append(np.mean(filtered))
    features.append(np.var(filtered))
    plt.subplot(1,4,i+1)
    plt.imshow(filtered, cmap='gray')
    plt.title(f"{int(np.degrees(theta))}°")
    plt.axis('off')
plt.suptitle("Material Texture Response")
plt.show()
print("Feature Vector:", features)
if np.mean(features) > 0.2:
    print("Predicted: Brick")
else:
    print("Predicted: Plaster / Wood")
```



https://github.com/Faiza-Khan109/